

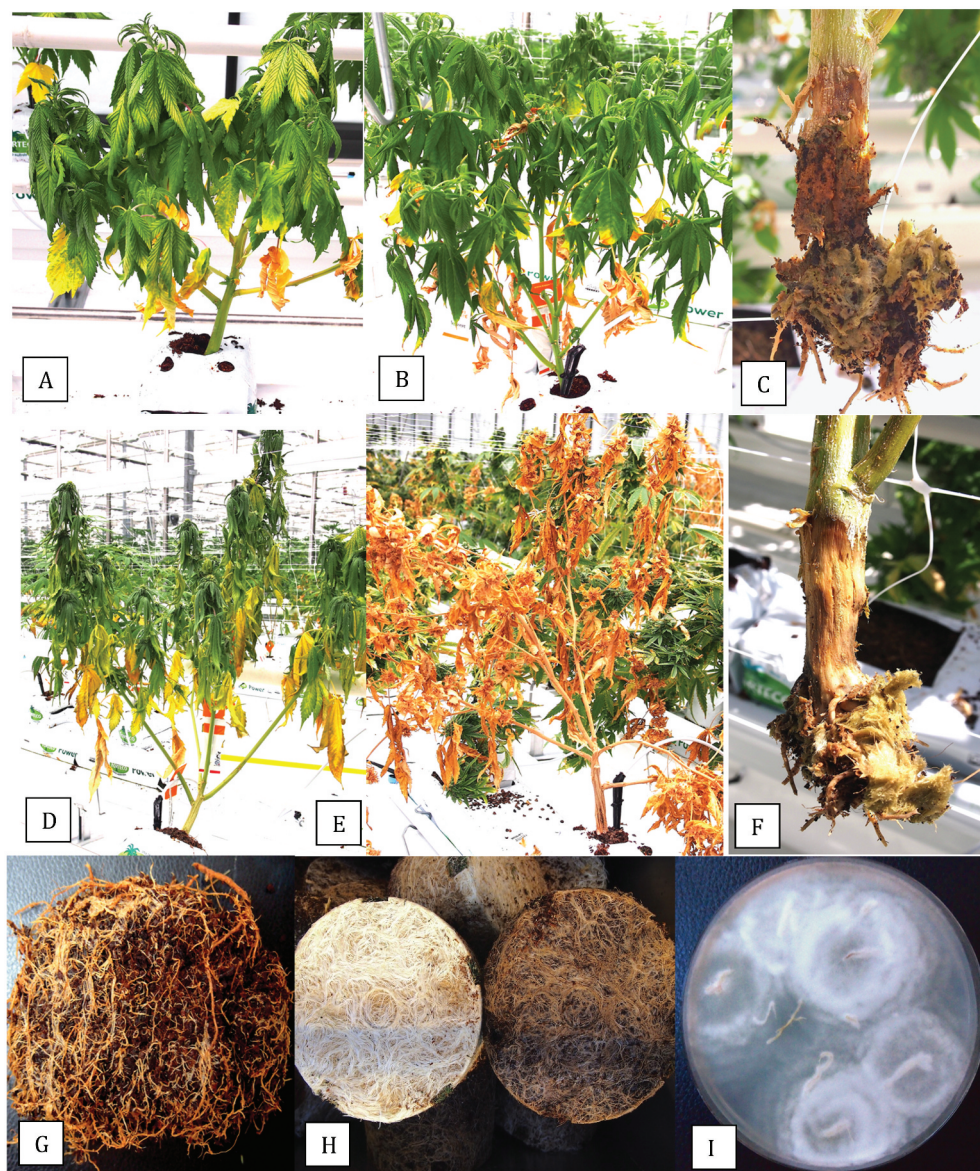


Fig. 2 (Colour online) Symptoms of root and crown rot on flowering plants of cannabis caused by several species of *Pythium*. **(a)** Early symptoms of wilting with darkening and yellowing of leaves in early stages of flowering. **(b)** Advanced stages of wilting on affected plant. **(c)** Crown decay and root destruction on plant shown in **(b)**. From this affected plant, *P. myriotylum* was recovered. **(d)** Extensive wilting and yellowing of leaves of plants at the flowering stage. **(e)** Total collapse and death of plants just prior to harvest. **(f)** Sunken crown lesion and destruction of the root system on plant shown in **(e)**. From this affected plant, *P. aphanidermatum* was isolated. **(g)** Browning of roots on flowering plants grown in a coco coir medium from which *P. dissotocum* was recovered. **(h)** Browning of roots on stock plant (right) compared to healthy plant (left). The affected plant had been subjected to long periods of water saturation. **(i)** Colony of *P. catenulatum* recovered from the affected root system shown in **(h)**.

Isolation from plants

Tissue sources for pathogen isolation included root segments, stem pieces and crown tissues from symptomatic plants. From each tissue type, small pieces, approximately 0.5 cm in length for roots or 0.2–0.4 cm² for cuttings and stem pieces, were surface-disinfested by

dipping them in a 10% bleach solution (containing 0.625% NaOCl) for 1 min, followed by 30 s in 70% EtOH. They were rinsed thrice in sterile water and blotted on sterile paper towels. Tissue pieces were plated onto potato dextrose agar (PDA, Sigma Chemicals, St. Louis, MO) amended with 130 mg L⁻¹ of streptomycin sulphate (PDA+S). Dishes were incubated under